

CH 412: Statistical Mechanics and Kinetics

Tutorials and Homeworks

Recapitulation of Thermodynamics

Tutorial (01/09/26)

1. Evaluate the following partial derivatives.

a) $P(V, T) = \frac{nRT}{V}$; $\left(\frac{\partial P}{\partial V}\right)_T$

b) $\left(P + \frac{a}{V^2}\right)(V - b) = nRT$; $\left(\frac{\partial P}{\partial V}\right)_T$ and $\left(\frac{\partial P}{\partial T}\right)_V$

2. Consider the internal energy $U = TS - PV + c$ for constant N , where c is an arbitrary constant.

(a) Write the total differential of U :

Hint: $z = xy \implies dz = xdy + ydx$; and $z = x + y \implies dz = dx + dy$

(b) Comparing with the total differential of U , we learned from the postulates, show that:

$$-SdT + VdP = 0$$

3. For the exact differential

$$df(x, y) = a(x, y)dx + b(x, y)dy$$

the following identity follows from Euler's test

$$\left(\frac{\partial a}{\partial y}\right)_x = \left(\frac{\partial b}{\partial x}\right)_y$$

Now consider the thermodynamic relation

(a) Starting from $dH(S, P) = TdS + VdP$ complete the analogous Maxwell relation

$$\left(\frac{\partial V}{\partial S}\right)_P =$$

(b) Starting from $dA(T, V) = -SdT - PdV$ complete the Maxwell relation

$$\left(\frac{\partial S}{\partial V}\right)_T =$$

(c) Starting from $dG(T, P) = -SdT + VdP$ complete the Maxwell relation

$$\left(\frac{\partial V}{\partial T}\right)_P =$$

Homework 1

(Due: 01/16/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

1. (5 + 5 points) Evaluate the following partial derivatives.

a) $\rho(P, T) = \frac{PM}{RT}$; $\left(\frac{\partial \rho}{\partial T}\right)_P$

b) $q(E_1, E_2, T) = e^{-E_1/kT} + e^{-E_2/kT}$; $\left(\frac{\partial q}{\partial E_1}\right)_{E_2, T}$

2. (5 points) A certain gas obeys the equation of state

$$P(V - nb) = nRT$$

where n and R are constants. Determine the coefficient of thermal expansion

$$\alpha = \left(\frac{1}{V}\right) \left(\frac{\partial V}{\partial T}\right)_P$$

3. (10 points) Find the expressions for T , P , and μ by partial differentiation of the internal energy given by:

$$U(S, V, N) = \left(\frac{v_o^2 \theta}{R^3}\right) \frac{S^4}{NV^2}$$

The v_o , θ , and R are constants.

4. (5 points) If the following relationship holds for a function $f = f(x, y, z)$:

$$df = -kdx + tdy + wdz$$

find the values of the partial derivatives:

$$\left(\frac{\partial f}{\partial x}\right)_{y,z}, \left(\frac{\partial f}{\partial y}\right)_{x,z}, \text{ and } \left(\frac{\partial f}{\partial z}\right)_{x,y}$$

5. (10 points) Consider the total differential:

$$dU(S, V, N) = TdS - PdV + \mu dN$$

Further consider the function $G = \mu N = U - TS + PV$. Using your knowledge of exact differentials, derive an expression for dG and substitute the result for dU to prove that:

$$-SdT + VdP - Nd\mu = 0$$

This result is known as the Gibbs-Duhem equation.

Recapitulation of Thermodynamics - II

Tutorial (01/16/26)

1. Show that

$$\mu = \left(\frac{\partial A}{\partial N} \right)_{V,T} \text{ and } V = \left(\frac{\partial G}{\partial P} \right)_{N,T}$$

2. (a) Starting from $U = U(S, V, N)$, generate a function $H = H(S, P, N)$ using legendre transformation.
(b) Write an expression for the differential of H , dH , and from this derive the expressions for the partial derivatives of H with respect to S , P , and N .
(c) The function H is known as enthalpy. Derive a relationship between enthalpy and Gibbs' Free Energy, which we discussed in class.
3. For the flipping of a fair coin, we express the probability of getting n heads out of N coin flips as $p(n|N)$. Compute the following quantities:
(a) $p(5|10)$
(b) $p(5|10)/p(4|10)$
(c) $p(10|20)/p(9|20)$.

Do you observe that the likelihood of observing 50% heads relative to slightly less than 50% heads increases with the total number of coin tosses?

(This question will need the use of a calculator.)

Homework 2

(Due: 01/23/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

1. (6 + 4 points) (a) Starting from $U = U(S, V, N)$, perform sequential Legendre transforms to derive an expression for the grand potential $\Xi = \Xi(T, P, \mu)$, which is a function of only intensive variables.
(b) Find out the partial derivatives of the grand potential with respect to all three intensive variables.

2. (5 + 5 Points) From the relationship $G = H - TS$, prove the following partial derivative relations:

$$\left(\frac{\partial(G/T)}{\partial T} \right)_P = -\frac{H}{T^2} \quad \text{and} \quad \left(\frac{\partial(G/T)}{\partial(1/T)} \right)_P = H$$

Note: these relationships are known as Gibbs-Helmholtz equations.

3. (12 points) Starting from

$$dU = TdS - PdV + \mu dN,$$

show that at constant T , P , and N , the maximization of the entropy of the universe (S_{univ}) is equivalent to minimizing the Gibbs Free Energy of the system (G_{sys}).

Hint: Assume the volume of the universe is constant:

$$dV_{\text{univ}} = 0 \implies dV_{\text{sys}} + dV_{\text{surr}} = 0$$

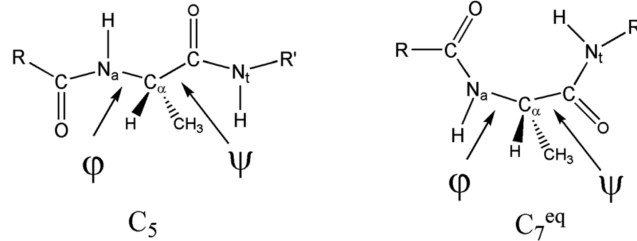
4. (4 + 4 points) Let us assume a molecule can remain in three different conformations with energies: E_1 , E_2 , and E_3 . Let's say the number of molecules in the energy states is n_1 , n_2 , and n_3 .
(a) Write the expressions for the probabilities p_1 , p_2 , and p_3 , of being in states 1, 2, and 3, respectively.
(b) Show that the average energy per molecule is

$$\langle E \rangle = p_1 E_1 + p_2 E_2 + p_3 E_3$$

Probability and Its Relation to Entropy

Tutorial (01/23/26)

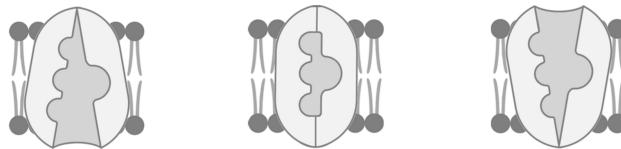
1. Alanine dipeptide is a molecule often used as a model system for testing statistical mechanical theories. In aqueous solution, it primarily remains in two states: $C7_{eq}$ and $C5$. These two conformations are equally likely.



- (a) If there are N alanine dipeptides in a solution, calculate the number of microstates for the macrostate where n of them are in C_5 conformation.
- (b) Compute the Boltzmann entropy for this macrostate, where n molecules are in C_5 conformation.
- (c) Consider this solution to be an isolated system, where you can find the equilibrium state by just maximizing the entropy. (We do not need free energy as there is no energy exchange with surrounding.) Find out at which value of n the entropy is maximum by setting $\frac{dS}{dn} = 0$. How many of the alanine dipeptides will be in C_5 conformation at equilibrium?

Hint: Use Sterling's approximation without the square root factor.

2. Deoxyribonucleic acids (DNA) are made of four base pairs, namely Adenine (A), Thymine (T), Guanine (G), and Cytosine (C). During an unbiased construction of a DNA, these four base pairs are equally likely to occur.
- (a) In a DNA octamer (8 bases), what are the probabilities of finding the sequence d(GCGCACGT) by random chance?
- (b) What is the probability of having exactly 3A, 3T, 2G, and no C in an octamer sequence?
- (c) Compute the Boltzmann entropy for the sequences in (b).
3. The NKCC1 membrane transporter protein transports Cl^- ions across cell membrane. This protein can remain in three conformations: inward open (IO), closed (C), and outward open (OO). All three conformations are equally likely.



Consider that in a cell, there are exactly 3 NKCC1 transporters that operate independently of one another.

- (a) Out of the three proteins, how many microstates (such as {OO,IO,OO}) are possible?
- (b) Compute the Shannon Entropy of the following macrostates: (i) [three C], (ii) [one IO and two OO], and (iii) [one IO, one OO, and one C].
- (c) Which of these three macrostates is most probable based on maximum entropy?

Homework 3

(Due: 01/30/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

1. There are 20 naturally occurring amino acids that are found in proteins.
 - (a) (4 points) Find out, in a 10-residue-long polypeptide, how many distinct sequences can occur given that a particular amino acid can be repeated more than once.
 - (b) (4 points) What is the probability that all ten amino acids are identical?
 - (c) (4 points) Calculate the Boltzmann Entropy of a macrostate where three residues are lysine, three arginine, and four are alanine.
 - (d) (4 points) Calculate the probability of such an occurrence as in (c).

2. Consider a lattice with N sites and n gray particles. Consider another lattice, adjacent to the first, with M sites and m black particles. Each site can hold only 1 particle. The number of lattice sites in the picture below is only a representation. **Do not** plug in numerical values of N , M , n , and m based on the figure below.



- (a) (4 points) Assume that the gray and black particles cannot switch lattices. This is state A. What is the total number of configurations W_A of the system in state A?
- (b) (4 points) Now assume that all $N + M$ sites are available to all the particles. The particles remain distinguishable by their color. This is state B. Now, what is the total number of configurations W_B of the system?

Now take $N = M$ and $n = m$ for the following problems.

- (c) (4 points) What is the ratio W_A/W_B ?
- (d) (8 points) Using Stirling's approximation (without the square root term), calculate the entropy difference given by

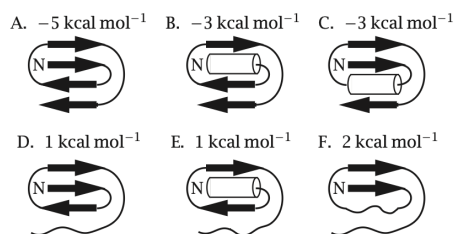
$$\Delta S = S_A - S_B = k_B \ln \left(\frac{W_A}{W_B} \right)$$

- (e) (4 points) Which state, A or B, has the greater entropy? What can you infer about the entropy of mixing from this exercise?

Boltzman Probability and Partition Function

Tutorial (01/30/26)

1. Assume a protein has six different discrete conformations, with energies given in the figure below:

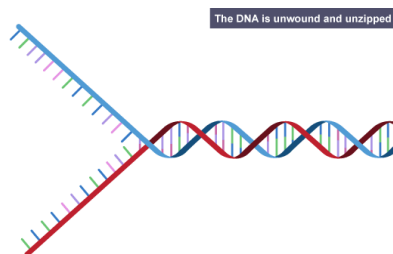


Calculate the following Boltzmann probabilities: (use $1 k_B T = 0.6 \text{ kcal/mol}$.)

- (a) $p(\text{state A})$
 (b) $p(1 \text{ kcal/mol})$
2. Find a relationship between the U and Q in terms of the temperature T instead of β , according to the chain rule of derivatives.

$$U = -\frac{\partial \ln Q}{\partial \beta}$$

3. In the process of DNA unzipping, the hydrogen bonds between the complementary nucleotides are broken, and the two strands become separated. Consider a system consisting of DNA with N base pairs. A broken base pair has a higher energy ($E = \varepsilon$) than a paired one ($E = 0$). This is because hydrogen bonds stabilize the system and reduce the energy. Also, for the n -th base pair to remain open, the previous ($n - 1$) base pairs should be open. (Note: for this problem, assume that the DNA can be unzipped only from one end.)



- (a) Calculate the partition function for this system. You need to use the expression:

$$\sum_{i=0}^N x^i = \frac{1 - x^{N+1}}{1 - x}$$

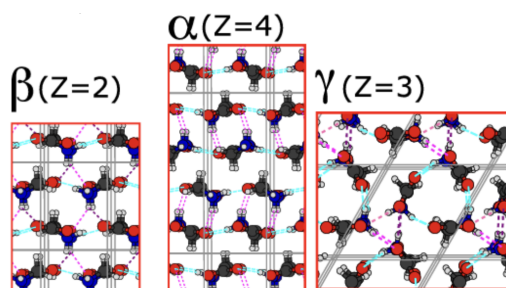
- (b) Find an expression for average energy $\langle E \rangle$, from which obtain the expression for $\langle n \rangle$, where n is the number of open base pairs.
 (c) What is the value of $\langle n \rangle$ at $T \rightarrow 0$? What is the physical meaning of this?

Homework 4

(Due: 02/06/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

1. You have a thermodynamic system with three states with energies ϵ_1 , ϵ_2 , and ϵ_3 . Assume $\epsilon_1 = 0$.
 - (a) (4 points) Write down the partition function for this system.
 - (b) ($4 \times 3 = 12$ points) What is the average energy, entropy, and Helmholtz free energy for this system?
 - (c) (4 points) If the probabilities of the three states are found to be $p_1 = 0.9$, $p_2 = 0.09$, and $p_3 = 0.01$ at $k_B T = 1$, what are the values of ϵ_2 and ϵ_3 ?
2. (4 points) Using the partial derivative of Helmholtz free energy, derive an expression for computing pressure in terms of the partition function.
3. Crystals of organic molecules often exhibit polymorphism, i.e., the same molecule can exist in different crystalline forms called polymorphs. Below is a picture of the three most commonly found polymorphs of glycine crystal: α , β , and γ .



The energies of different states are $E_\alpha = 3\epsilon$, $E_\beta = 5\epsilon$, and $E_\gamma = 0$. The degeneracy of each polymorph is one.

- (a) (5 points) Compute the partition function for glycine polymorphism and compute the fraction of each polymorph.
 - (b) (5 points) Compute the heat capacity (C_V) of a glycine crystal with only these three states accessible to it. Plot qualitatively C_V vs T . You can use plotting software, but draw the plot by hand in the homework with appropriate axis labels.
- (Hint: First calculate $\langle E \rangle$ from Q , then use $C_V = \partial \langle E \rangle / \partial T$. Do not use $C_V = \sigma_E^2 / k_B T$).
4. (This problem may be easier to solve after the class on Monday/Tuesday that covers Polymer folding)

A polymer can remain in folded, misfolded, or unfolded conformations. The energies and degeneracies of these states are: E_{folded} , $E_{\text{misfolded}}$, and E_{unfolded} ; and g_{folded} , $g_{\text{misfolded}}$, and g_{unfolded} , respectively. Assume $E_{\text{folded}} = 0$

[Interactive Tutorial](#)

Using this interactive tutorial above, determine how the melting point (T_m) of the polymer will change if we

- (a) (2 points) Increase the degeneracy of the folded state while keeping everything else constant.
- (b) (2 points) Increase the energy of the unfolded state, keeping everything else constant.
- (c) (2 points) Melting point is the temperature at which the probabilities of the states follow this relation:

$$p(\text{folded}) = p(\text{misfolded}) + p(\text{unfolded})$$

Based on this relationship, explain the physical meaning of your observation in (a) and (b).

Boltzman Probability and Partition Function

Tutorial (02/06/26)

1. Consider a system of N particles. Each particle can occupy only three discrete states of energies 0 , ε , and 2ε . The system is kept at a fixed temperature T . Write the partition function of the system, considering
 - (a) the particles are *distinguishable*
 - (b) the particles are *indistinguishable*
 - (c) Derive an expression for the entropy (S) and heat capacity (C_V) of the system at constant volume.
2. In its biologically active state, a protein always resides in the same native folded state. The unfolded state, however, can be a multitude of states. Let us consider that there are N such unfolded states with energy $E = 0$ and there is a single unique folded state with energy $-\varepsilon$. Note that the folded state has lower energy than the unfolded state due to intramolecular interactions such as hydrogen bonds.
 - (a) Write down the partition function of this system.
 - (b) At the melting temperature T_m , the probability of being in the folded state is equal to that of being in the unfolded state. Derive an expression for T_m .
 - (c) At any generic temperature T , find the free energy difference between the folded and unfolded states: $\Delta A = A_F - A_U$. Show that ΔA changes sign at $T = T_m$.

Molecular Partition Function

Tutorial (02/16/26)

1. Review midterm problems as necessary.
2. (a) Derive an expression for the probability of finding a 1D harmonic oscillator in its ground state, i.e., the lowest energy state.
(b) From your result in (a) find the expression for the temperature at which exactly half of the population is in the ground state.
3. (a) Derive an expression for the most populated energy level of a rigid rotor.
(b) What is the most probable value of the rotational quantum number J of the gas phase N_2 molecule at 298 K? Moment of inertia of N_2 is $2.51 \times 10^{-47} \text{ kg} \cdot \text{m}^2$

Homework 5

(Due: 02/20/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

- Consider an RNA aptamer system that can exist in two microstates: Unbound (u) and Bound (b). Let the energies of these states be E_u and E_b , respectively. The system is in thermal equilibrium with a reservoir at temperature T .

a) (3 points) Show that the total Helmholtz free energy A of the system can be expressed as:

$$A = E_u - \frac{1}{\beta} \ln \left(1 + e^{-\beta(E_b - E_u)} \right)$$

where $\beta = \frac{1}{k_B T}$.

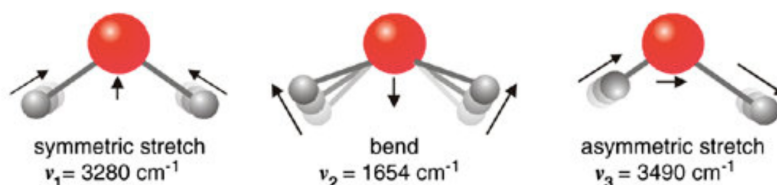
b) (4 points) Derive the expression for the probability of finding the RNA in the bound state, p_b . Discuss the limit of p_b as $T \rightarrow 0$ and $T \rightarrow \infty$ for the case where $E_b < E_u$.

Consider $\Delta E = E_b - E_u$. Compare two ligands, Theophylline ($\Delta E = -\epsilon$) and Caffeine ($\Delta E = \epsilon$). At a temperature such that $\beta\epsilon = 1$:

c) (4 points) Calculate p_b and p_u for both ligands.

d) (3 points) Determine which drug is a more effective binder based on the sign of the Free Energy change $\Delta A_{bind} = A_b - A_u$.

- (5 points) Derive the expression for frequency-dependent entropy for a single particle one-dimensional harmonic oscillator system. Simplify the expression to the extent possible by cancelling off redundant terms.
- The bond stretching frequency of an N_2 molecule is $7 \times 10^{13} \text{ s}^{-1}$. The values of Planck's constant and Boltzmann constant are $h = 6.626 \times 10^{-34} \text{ Js}$ and $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$, respectively.
 - (4 points) Calculate the ratio of zero-point energies and thermal energy ($k_B T$) at 300 K.
 - (4 points) At which temperature are the zero-point energy and thermal energy equal?
- The three different vibrational modes of water and their corresponding frequencies are depicted below. Consider these modes constitute three independent harmonic oscillators.



(a) (5 points) Calculate the vibrational partition function of a single water molecule. Use $h = 6.626 \times 10^{-34} \text{ Js}$, $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$, and $1 \text{ cm}^{-1} = 3 \times 10^{10} \text{ s}^{-1}$.

(b) (4 points) Without substituting the numerical values (i.e., just using ν_1 , ν_2 , and ν_3), show that the vibrational heat capacity for such a system is the sum of the heat capacities of each individual oscillator.

(c) (4 points) Using your result from (b), calculate the vibrational heat capacity of a single water molecule at 300 K temperature.

Homework 6

(Due: 02/27/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

1. The quantum mechanical energy levels of a particle in a 2D box of sides a are given by

$$E_{n_x, n_y} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2)$$

- (a) (8 points) Derive the translational partition function of this system in the classical limit. Use

$$\int_0^\infty e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}$$

- (b) (6 points) Find also the expressions for the internal energy and heat capacity.

2. (8 points) Starting from the partition function, show that the entropy of a 3D monoatomic ideal gas consisting of N particles in a container of volume V and temperature T is given by

$$S = Nk_B \left[\ln \left(\frac{V}{N} \right) + \frac{3}{2} \ln \left(\frac{2\pi mk_B T}{h^2} \right) + \frac{5}{2} \right]$$

3. Consider a system with two compartments with an impenetrable partition. Both compartments of equal volume V are filled with the same 3D monoatomic ideal gas of N molecules, each with total energy E . The whole system is in thermodynamic equilibrium.

(a) (10 points) Considering the gas particles are *distinguishable*, calculate the change in entropy ΔS after removing the partition, when the gases are allowed to mix.

(b) (4 points) Repeat the calculation considering the gas particles are *indistinguishable*. You can use the expression of entropy in problem 2. Show that this process only results in an increase of entropy when the gas particles are *distinguishable*.

4. (4 points) Qualitatively draw the C_V vs T plots for Ar and Cl_2 gas across the whole temperature range and explain the reasons behind any observed difference between them.

Concept of Ensembles and Chemical Equilibrium

Tutorial (02/27/26)

1. For an ideal gas the canonical partition function is

$$Q_N(T, V) = \frac{1}{N!} \left(\frac{V}{\Lambda^3} \right)^N$$

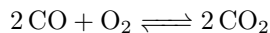
where Λ is the thermal De-Broglie wavelength.

Calculate the grand canonical partition function for an ideal gas where the number of particles can go up to ∞ . Obtain a compact expression without any factorial or summation.

Hint: Use

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = e^x$$

2. The binding affinities of three drug candidates are $3.2 M$, $0.019 M$, and $0.03 mM$. What are their binding free energies in kcal/mol at 300 K? ($RT = 0.6$ kcal/mol). In the absence of any other information, which drug candidate is likely to be most effective? Is any of them a “non-binder”?
3. Consider the following reaction:

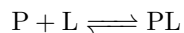


- (a) Write an Expression for the equilibrium constant K_{eq} of this reaction in terms of the partition function of each molecule and their ground state energies: $E_0(\text{CO})$, $E_0(\text{O}_2)$, and $E_0(\text{CO}_2)$.
- (b) You performed an experiment where you measured the K_{eq} of this reaction as a function of temperature. Write the expressions for the slope and the y-intercept of the $\ln K_{eq}$ vs $1/T$ plot.
- (c) Calculate the value of the slope when the ground state energies of the molecules are $E_0(\text{CO}) = -533$ kJ/mol, $E_0(\text{O}_2) = -498$ kJ/mol, and $E_0(\text{CO}_2) = -799$ kJ/mol. (The energies are negative because they are relative to the energy of the individual atoms.)

Chemical Kinetics

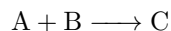
Tutorial (03/09/26)

1. Consider the problem of a protein ligand binding:



Consider that both the binding and unbinding processes are single-step elementary reactions.

- (a) What are the reaction orders of the binding and unbinding process?
 - (b) Sometimes, the ligand is added in so much excess that during the binding process, the ligand concentration does not change. What would be the reaction order of the binding process in that case?
 - (c) If the rate constant of binding is k_1 and dissociation k_2 , derive an expression for the binding free energy ΔG_b as a function of k_1 and k_2 .
2. For the following reaction:



the rate law is Rate: $d[C]/dt = k[A]^2$.

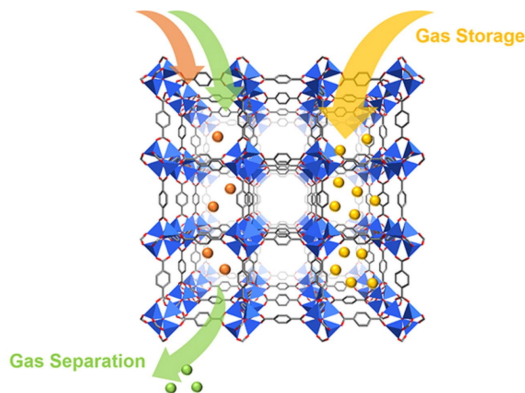
- (a) Which of the following mechanisms (involving elementary reaction steps) will be consistent with this rate law? M is an intermediate.
 - (i) Step 1: $A \xrightarrow{k_1} M$; Step 2: $M + B \xrightarrow{k_2} C$
 - (ii) Step 1: $A + B \xrightarrow{k_1} M$; Step 2: $M \xrightarrow{k_2} C$
 - (iii) Step 1: $A + A \xrightarrow{k_1} M$; Step 2: $M + B \xrightarrow{k_2} C$
 - (b) In the correct mechanism, what is the relationship between k_1 and k_2 ? Which step is the rate-determining step?
3. The rate of a chemical reaction doubles by increasing the temperature from $T_1 = 300$ K to $T_2 = 310$ K. What is the activation energy E_a for this reaction? Assume the pre-exponential factor is independent of temperature. Use $k_B = 2 \times 10^{-3}$ kcal s mol⁻¹.

Homework 7

(Due: 03/13/26 at 9 am in class)

Please only submit a handwritten paper copy of your work. Typed out or electronic copies are not accepted.

1. Metal Organic Frameworks (MOF) are molecular cage-like structures inside which gas molecules such as N_2 , CO_2 , etc., can be stored, and then released as necessary.



Consider an MOF with exactly 4 cavities, each of which can host one gas molecule. Each occupied cavity has energy $-\varepsilon$ and each empty cavity has energy 0. Assume the gas molecules do not interact with each other.

- (a) (4 points) Write an expression for the canonical partition function for this system.
 - (b) (8 points) Calculate the average internal energy from the canonical partition function.
 - (c) (4 points) Calculate the expression for the average number of particles ($\langle N \rangle$) adsorbed in this MOF using the result from part (b).
 - (d) (4 points) If we want to extract the stored gas by heating the MOF, the $\langle N \rangle$ should decrease with increasing temperature. Check whether this is true for this system. Hint: You can prepare a plot of $\langle N \rangle$ in Desmos and qualitatively draw the plot in hand to explain the effect.
2. (10 points) The rate constant of a chemical reaction can be expressed in terms of the internal energy of the activated complex (transition state) as:

$$k = B(T)e^{-\Delta U^\ddagger/k_B T}$$

where $\Delta U^\ddagger$ is the difference of the internal energy of the TS and the reactant. Considering that this reaction rate follows the Eyring-Polanyi equation, derive an expression for the pre-exponential factor $B(T)$.

3. (a) (5 points) The rate constant of an uncatalyzed reaction is $k = 10^3 \text{ s}^{-1}$. What is the Gibbs free energy of activation $\Delta G^\ddagger$ for this reaction? Use $k_B T = 0.6 \text{ kcal mol}^{-1}$ and Planck's constant $h = 10^{-13} \text{ kcal s mol}^{-1}$.
- (b) (5 points) An enzyme reduces the $\Delta G^\ddagger$ of the transition state by 5 kcal mol^{-1} . What is the rate constant of the enzyme-catalyzed reaction?